

Lab-4 Ex3 - Fibonacci Number Lookup

Wikipedia, "In mathematics, the Fibonacci numbers, commonly denoted F_n , form a sequence, the Fibonacci sequence, in which each number is the sum of the two preceding ones. The sequence commonly starts from 0 and 1, although some authors omit the initial terms and start the sequence from 1 and 1 or from 1 and 2. Starting from 0 and 1, the next few values in the sequence are: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, ...".

We can write the relations as $F_n = F_{n-1} + F_{n-2}$.

In the lecture, we have learned that the array can be served as a lookup table. Use the lookup table to output the Fibonacci number based on the input value.

Input:

An integer n indicating the n -th Fibonacci number needed.

You can assume n ranges between 0 and 20 inclusively.

Output:

An integer F_n .

For example, if $n = 0$, then $F_0 = 0$.

If $n = 1$, then $F_1 = 1$.

If $n = 2$, then $F_2 = 1$.

... etc

User-input integers are ***bold and italic*** in the following sample runs.

Sample Run #1:

0

0

Sample Run #2:

19

4181